

The Ephemeris

March 2018

Volume 29 Number 01 - The Official Publication of the San Jose Astronomical Association



Mar 2018 - June 2018 Events

Board & General Meetings

Saturday 3/3, 3/31, 4/28, 5/26

Board Meetings: 6 -7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 3/4, 4/8, 5/6, 6/3

Solar Observing (locations differ)

Sunday 3/4, 4/21, 5/6, 6/3

Intro to the Night Sky Class

Houge Park 1Q In-Town Star Party

Friday 3/23, 4/20, 5/18, 6/22

Houge Park 3Q In-Town Star Party

Friday 3/9, 4/6, 5/4, 6/8

RCDO Starry Nights Star Party

Saturday 3/10, 4/7, 6/9

Imaging SIG Mtg

Tuesday 3/20, 4/17, 5/15, 6/19

Astro Imaging Clinics

(at Coyote Valley OSP)

Saturday 5/5

Quick STARt (by appointment)

Friday

Binocular Star Gazing

6/9

Please refer to the SJAA Website home page (www.sjaa.net) and Click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancelation due to weather.

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January 31, 2018 Lunar Eclipse

Location: Sunnyvale, California

Photo: SJAA Member: **Jordan Makower**

SJAA Annual Awards and Potluck Feb 3, 2018 by Rob Jaworski

This year's annual Membership Meeting of the San Jose Astronomical Association was held on the evening of Saturday, February 3, 2018. It was a night of gain and loss, with recognition and camaraderie.

First off, the business: board elections were held, for four of the nine seats. Incumbents Bill O'Neil, Rob Jaworski, Wolf Witt and Glenn Newell were all near unanimously re-elected for another two year term. There were no write-in candidates.

At this event, we celebrate those who have stepped up to donate the most important of assets, their time, to furthering the mission of the SJAA. This year saw four club members recognized for their contributions with Certifications of Recognition, as well as a small bouquet of flowers from master gardener and SJAA member Marianne Damon.

First was David Grover, whose monthly Intro to the Night Sky introduces families, kids and everybody to what is up in the sky in the current month. Every month on the Friday closest to the first quarter moon, David hosts the one-hour talk to explain what can be seen from the average bay area back yard. Afterward, participants are able to go outside and attend the In Town Star Party, where astronomer volunteers are set up with telescopes to show those same objects and concepts. Next to the School Star Party program, David's Intro to the Night Sky series does a lot to fulfill the SJAA's mission of bringing astronomy and science to the public.

Second was Glenn Newell, who runs the SJAA's Hands On Imaging program. Glenn has been a board member for several years and has taken his interest in astrophotography to a new level, setting up and running the Hands On Imaging program. This program allows members and the public to come out to darker skies, try their hand (and their rig) at capturing images of the night sky under supervision of others who might know a little bit more about the topic. It's a safe place to give this most precise aspect of amateur astronomy a try. Glenn has also been a reliable volunteer at swap meets and other SJAA events.

Next was Sukhada Palav, the SJAA's first librarian. Sukhada ran with the idea of setting up a library of physical books, taking in donations, identifying and procuring suggested volumes for the library, and cataloging and indexing the collection. She has also had to make decisions on what stays and what goes; indeed, though most librarians are bibliophiles, the honest sort will recognize that some texts have outlived their usefulness and culling of the herd is necessary. Sukhada has also been running the guest speaker program over the past eighteen months. She has done an amazing job of bringing in all types of speakers, from post docs and professional astronomers to amateur eclipse chasers telling their stories. She goes so far as to organize the informal annual show-and-tell (known as the Slide and Equipment night for the long-timers) and other programs for the monthly general meetings.

On behalf of the SJAA Board of Directors, to Sukhada, Glenn, and David, a huge THANK YOU for your generous commitment of time and energy for the benefit of our astronomy community.

At the meeting we also had some less happy news. We learned that the current SJAA president, Teruo Utsumi, has experienced recent health issues and doesn't yet have the energy needed to meet the demands of the job. As a result, he tendered his resignation from both the president role as well as the board of directors. Teruo has been a member of the SJAA for many years and had contributed in many other ways outside of being a long time board member. He has run the speaker program, served for years in the executive position of secretary, and has contributed greatly to the School Star Party program and other outreach events. For all this, the SJAA board of directors has also celebrated Teruo's contributions with a Certificate of Recognition. The board thanks Teruo for his services, hopes he gets back to his vibrant self everyone knows, and looks forward to seeing him back under the stars with other SJAA members very soon.

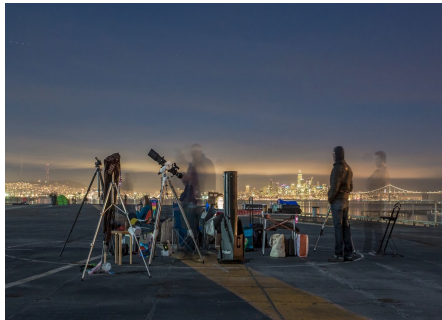
The evening went on with the SJAA gaining a couple of new members and a wonderful potluck to which the twenty five attendees contributed and enjoyed. Vini's and Leslie's homemade lasagna was delicious, and Marianne once again did a wonderful job with the setting. Flowers and candles may have been a first at an SJAA potluck!



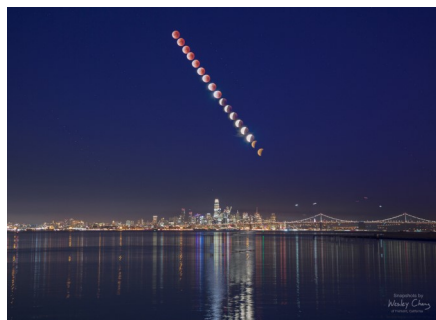
Above Vice President Bill O'Neil gives certificates to Glenn Newell and Sukhada Palav and the flowers from Marianne Damon

Lunar Eclipse from the USS Hornet - Alameda, CA

Report and photos by Wesley Chang from Eastbay Astronomy Society



*A small contingent of moon gazers assembling near the center of the deck with telescopes and binoculars
Photo courtesy of Wesley Chang*



*A series of photos tracking the movement and changing visage of the moon
Photo courtesy of Wesley Chang.*



*Still partially eclipsed moon set right over Coit tower and the western span of the Bay Bridge
Photo courtesy of Wesley Chang*

Editor's Note: The SJAA members that attended this event are: Emil Nenov, Wolf Witt, Julia Yang and Peter Mu

First of all, THANK YOU to Bill O'Neil, VP at SJAA, for alerting us to the Eclipse event on the USS Hornet. It turned out to be a picture-perfect vantage point to watch the moon move through Earth's shadow and descend over the San Francisco skyline.

The ship opened up around 7pm, and I along with Ray Wong and Tony Flusche joined about 150 or so overnight attendees. We all shuffled up the gangway and on to the ship, carrying our cameras, binoculars, and sleeping bags. After checking into our bunks near the ship's forecabin and dropping off our belongings, we quickly climbed to the upper deck to check out the vistas around us. Among the visitors on deck were a handful of TV news crew reporting on the impending lunar eclipse. I was actually able to watch the KPIX reporter give a live report during the 10 PM newscast. There was also a small contingent of moon gazers assembling near the center of the deck with telescopes and binoculars. Star party aboard an aircraft carrier is a "first" for me. :)

Admittedly, the USS Hornet is not suitable for "standard" stargazing. Light pollution for nearby San Francisco and Oakland, combined with the bright pier lights obscure all but the brightest stars. Standing on the deck, I could make out Orion, but I had difficulty locating Cassiopeia and even Polaris. But of course, the full moon was out, and she was going to be the star of the show throughout the evening and following morning, so we were not looking for dark skies at all. For me the main objective for coming out to Alameda and specifically the USS Hornet was to be able to watch and photograph the eclipsed moon descend over the San Francisco skyline. Even before learning about the overnight event on the USS Hornet, I had intended to come to the adjacent pier with Ray and Tony. (I had chosen this site after playing around with www.mooncalc.org to determine the azimuthal position of the moon.)

I had never visited the USS Hornet Museum, so after touring the top deck, we spent much the evening hours exploring the ship. As most of you probably know, in addition to its storied history dating back to World War II, the USS Hornet played an important role in the U.S. Space Program: the Hornet was the ship that picked up the crews of Apollo 11 and Apollo 12 after splashdown. Among the exhibits on board are a Block I Apollo Command Module and an actual Mobile Quarantine Facility, though it was the one used by the Apollo 14 crew. In this very hanger, President Nixon welcomed the Apollo 11 back to Earth after their historic mission. (Hornet + 3). So even though we didn't do much "stargazing", we did get to see a piece of "space" history.

The time passed quickly, and soon the 11 PM hour was upon us. It was time for "lights out", and we were all instructed to return to our bunks. We each crawled our way into our sleeping berths to catch a few ZZZ's before the Big Show. But just a few hours later, since the umbral eclipse began at 3:48 AM, we were all roused from our brief slumber at exactly 3:15 AM, Navy style, with whistle and "Now hear this..." announcements. We quickly jumped out from our berths, grabbed cameras and binoculars and made our way to the aft of the ship to stake out a spot with an unobstructed view of the western sky. I was surprised to find that on the adjacent pier below, there was already a hoard of photographers crowded at the end, all vying for a vantage point. I was glad to be aboard the ship. And even though I was sleep-deprived, the excitement of the event kept me wide awake through the whole morning.

After the total phase of the eclipse, as the umbral shadow receded, and the sky began to brighten from the approaching dawn, the moon eventually set over the City. The vista of a partially eclipsed moon setting over the familiar landmarks of San Francisco during the pre-dawn twilight was another spectacular treat to witness. As a finale, a still partially eclipsed moon set right over Coit tower and the western span of the Bay Bridge, just as the sun started to come up on the opposite horizon. Fantastic show, from start to finish!

Epilogue. With weather cooperating, the eclipse viewing on board the USS Hornet took full advantage of the somewhat unusual characteristics of this particular lunar eclipse, namely the altitude and azimuthal position of the moon along with the timing of eclipse's end during the dawn twilight. The Hornet was ideally positioned for us to watch the eclipsed moon juxtapose against the San Francisco skyline and afforded numerous vantage points for most us spectators to have that "front row" seat. And, as with the solar eclipse last August, sharing the experience and that sense of awe with good friends and strangers alike made this event even more enjoyable and memorable.

Falcon Heavy: A Multi-User Spaceport Success



Lifting off at 3:45 p.m. from Launch Complex 39A at NASA's Kennedy Space Center, a SpaceX Falcon Heavy begins its demonstration flight. This is a significant milestone for the world's premier multi-user spaceport. In 2014, NASA signed a property agreement with SpaceX for the use and operation of the center's pad 39A, where the company has launched Falcon 9 rockets and prepared for the first Falcon Heavy. NASA also has Space Act Agreements in place with partners, such as SpaceX, to provide services needed to process and launch rockets and spacecraft.

Credits: NASA/Kim Shiflett

By Bob Granath

NASA's Kennedy Space Center, Florida

The launch of a SpaceX Falcon Heavy rocket on its demonstration flight is another sign that NASA's Kennedy Space Center in Florida is continuing to grow as the nation's premier, multi-user spaceport. The new vehicle lifted off from NASA's historic Launch Complex 39A at Kennedy at 3:45 p.m. EST on Feb. 6.

Acting NASA Administrator Robert Lightfoot congratulated the entire SpaceX team on the successful launch of the Falcon Heavy.

"All of us in this business know the effort it takes to get to a first flight of any new vehicle and recognize the tremendous accomplishment we witnessed today," he said. "I am really proud of the hard work of our NASA team, in particular at Kennedy, for the transformation into a multi-user spaceport. Watching the Falcon Heavy rise above the historic pad that has been the launch point for so many critical missions is a true testament to the hard work transitioning our nation's launch infrastructure in support of the commercial launch industry."

Kennedy Space Center Director Bob Cabana also expressed congratulations to Space X on the launch of the Falcon Heavy.



In a photograph taken Dec. 28, 2017, the first SpaceX Falcon Heavy stands at Launch Complex 39A during testing at NASA's Kennedy Space Center.

Credits: SpaceX

"The successful launch of a new vehicle on its first flight is a significant accomplishment they can be very proud of," he said. "As a multi-user spaceport, I look forward to the continued expansion of commercial spaceflight from Kennedy and the integration of a new class of launch vehicle into our Nation's space program."

In 2014, NASA signed a 20-year [property agreement with SpaceX](#) for the use and operation of the launch pad.

Beginning in 2011, Kennedy sought partnerships with the U.S. aerospace industry to use former space shuttle facilities. Today, NASA has partnerships with more than 90 companies that enable commercial space manufacturing, processing and launch operations along Florida's Space Coast.

(Continued on the following page)



Launch Complex 39A at NASA's Kennedy Space Center has undergone modifications by SpaceX to adapt to the needs of the company's Falcon 9 and Falcon Heavy rockets. A horizontal integration facility has been constructed for processing rockets prior to rolling out for liftoff. In 2014 NASA signed a property agreement with SpaceX for the use and operation of the historic pad used during the Apollo and Space Shuttle Programs.

Credits: NASA

(Falcon Heavy- continued from page 4)

Kennedy's first significant partnership with industry allowed Boeing to use Orbiter Processing Facility 3, now known as the Commercial Crew and Cargo Processing Facility, or C3PF. Here Boeing is manufacturing and processing its CST-100 Starliner spacecraft, which is slated to carry astronauts to the [International Space Station](#) for [NASA's Commercial Crew](#)

[Program](#). SpaceX will similarly launch astronauts from Launch Complex 39A for NASA.

A handful of NASA employees serve as customer advocates and support the space center's commercial partners. Through reimbursable Space Act Agreements, NASA provides services that are needed at a spaceport.

"We support launch campaign activities such as propellant and life support services, which includes providing commodities such as helium and nitrogen to support launch, flight hardware transport and roadblock security support near the launch pad," said Robyn Mitchell, the NASA customer advocate who supports SpaceX.

In her customer advocate role, Mitchell monitors the partner's processing and facilities schedules to help ensure their operations and mission milestones are met. This includes integrating support plans in response to the partner's requests for services.

"When partners, such as SpaceX, are preparing for a launch, we have Support Readiness Reviews," Mitchell said. "While SpaceX is responsible for the launch vehicle and payload, NASA organizations verify support requests are complete and confirm the readiness of Kennedy's facilities, equipment and infrastructure for launch."

Mitchell noted that the ongoing responsibilities of her office are key to establishing and maintaining a multi-user spaceport capability at Kennedy, specifically working with commercial partners to help them identify technical requirements, anticipate operational impacts and develop solutions taking advantage of the center's unique launch infrastructure and capabilities.

The test is evidence of the transformation of NASA's Kennedy Space Center as a multi-user spaceport. NASA and its commercial and international partners are looking to return humans to the Moon and beyond, and there will be opportunities for a wide range of rockets and capabilities.

NASA's [Space Launch System](#) (SLS) rocket and [Orion](#) spacecraft is targeted to lift off next year from nearby Launch Complex 39B. The initial SLS rocket is expected to have a liftoff thrust of 8.8 million pounds, and is designed to be an evolvable launch vehicle able to meet the most challenging deep space crew and cargo needs.

"As SpaceX continues to refine operations of its Falcon Heavy rocket, we welcome this capability to be added to those available soon from NASA's SLS," said Tom Engler, Kennedy's director of Center Planning and Development.

Rocket Girls and the Advent of the Space Age



In the earliest days of space exploration, most calculations for early space missions were done by “human computers,” and [most of these computers were women](#). These women's calculations helped the U.S. launch its first satellite, Explorer 1. This image from 1953, five years before the launch of Explorer 1, shows some of those women on the campus of the Jet Propulsion Laboratory (JPL).

JPL's human computers didn't just help launch the U.S. space program; they also represented an important step forward for women and other underrepresented people at a time when much of the professional world and especially technical fields were dominated by white men. Janez Lawson (seen in this photo), was the first African American hired into a technical position at JPL. Having graduated from UCLA with a bachelor's degree in chemical engineering, she later went on to have a successful career as a chemical engineer.

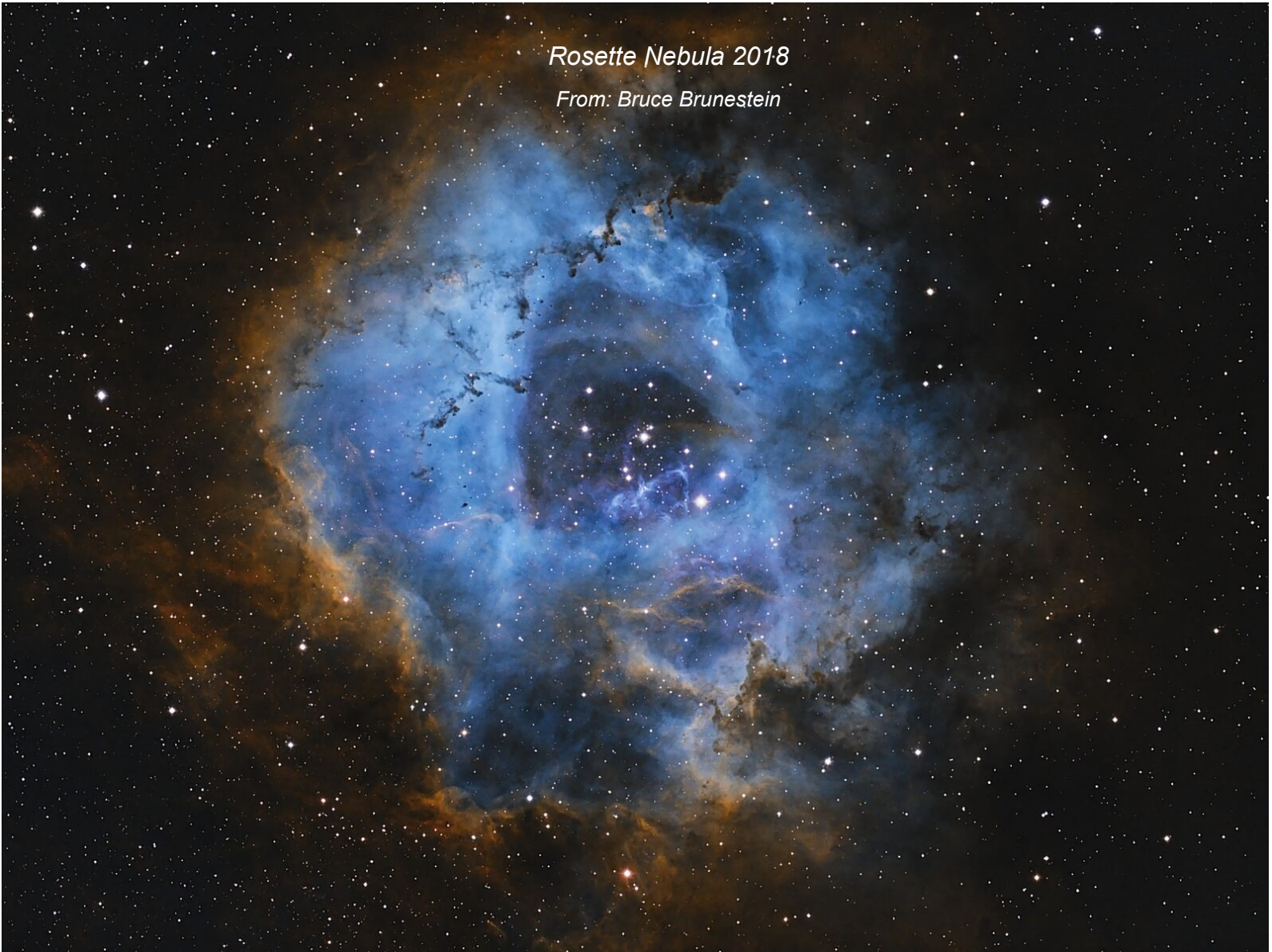
Explorer 1 successfully launched from Cape Canaveral's Pad 26 on January 31, 1958.

This image of JPL's human computers shows women—black and white, young and old—whose work was helped propel this nation into the space age. And they did so by hand.

Image Credit: NASA/JPL-Caltech

Rosette Nebula 2018

From: Bruce Brunestein



Location: Cambrian Park, San Jose, CA Feb 2nd—8th

Mount: Software Bisque Paramount MX+

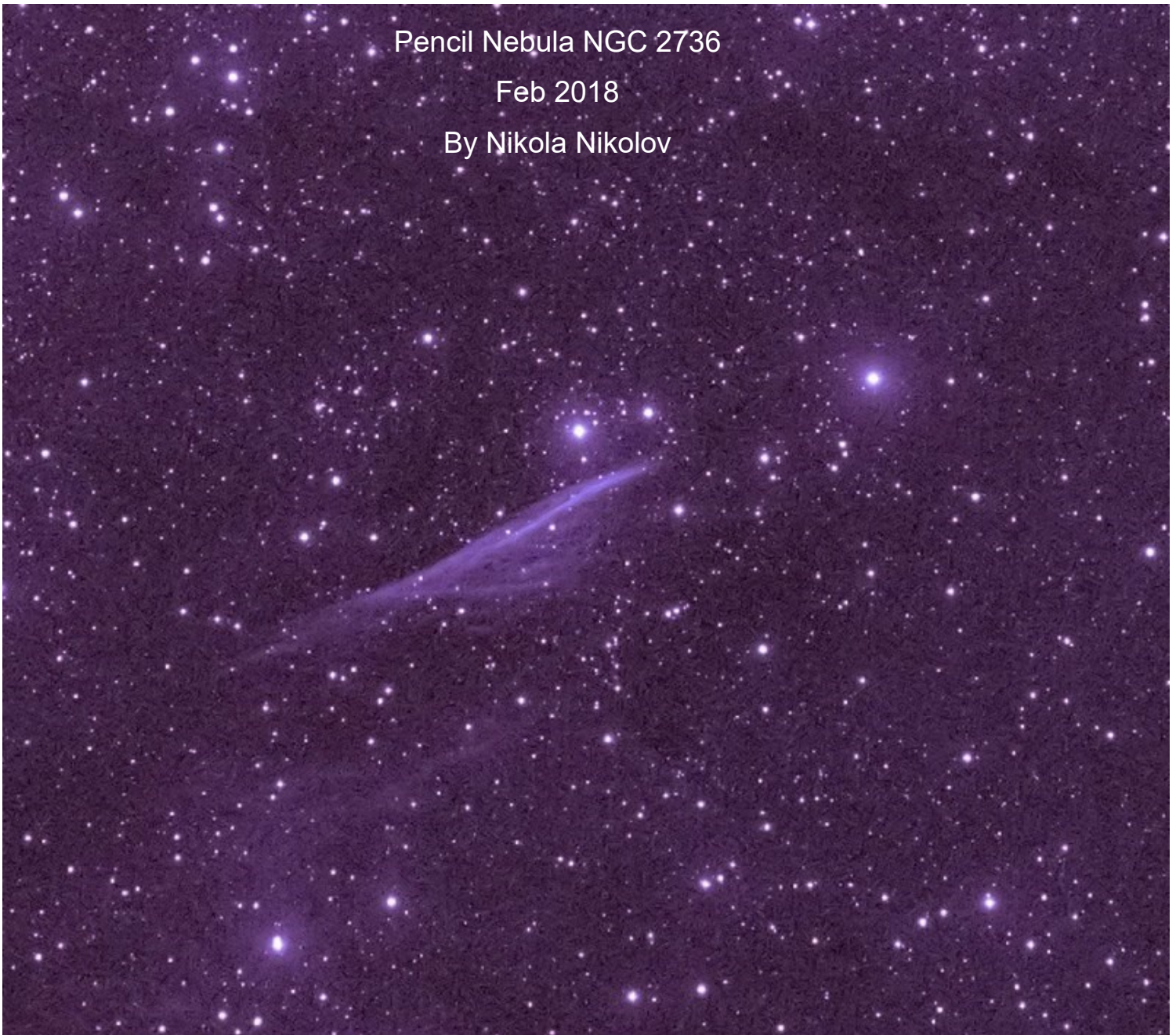
Camera: SBIG STT 8300M

*Exposure: SII 3nm 35x600" 1x1
Ha 3nm 28x600" 1x1*

Pencil Nebula NGC 2736

Feb 2018

By Nikola Nikolov



Telescope: iTelescope T12 Takahashi FSQ ED 106mm

CCD camera: FLI Microline 11002

Location: Siding Spring Observatory, Australia

Filters: Astrodon Lum, Red, Blue, Ha, SII

Total integration time: 1h

Exposure per frame: 300s

Frames: 3 x Ha, 3 x Blue, 5 x Red, 1 x Lum, 2 x SII

Processing software: Astro Pixel Processor, Photoshop

Melotte 15 by Alan Pham



AP ALAN PHAM
ASTROPHOTOGRAPHY

Imaging telescope or lens: [Stellarvue SV4](#)

Imaging camera: [ZWO ASI1600MM-Cool](#)

Mount: [Astro-Physics Mach1 GTO](#)

Guiding telescope or lens: [Stellarvue SV4](#)

Guiding camera: [QHY5L-II-M](#)

Focal reducer: [Astro-Tech AT2FF Field Flatteners](#)

Software: [PS](#), [Straton](#), [PixInsight](#), [Sequence Generator Pro](#), [PHD2](#)

Filters: [Astrodon SII 1.25 3nm](#), [Astrodon OIII 1.25 3nm](#), [Astrodon Ha 1.25 5nm](#)

Resolution: 2500x1459

Dates: [Oct. 21, 2017](#), [Nov. 17, 2017](#), [Dec. 21, 2017](#)

Frames:

[Astrodon Ha 1.25 5nm](#): 215x180" (gain: 200.00) -15C bin 1x1

[Astrodon OIII 1.25 3nm](#): 179x180" (gain: 200.00) -15C bin 1x1

[Astrodon SII 1.25 3nm](#): 161x180" (gain: 200.00) -15C bin 1x1

Location: San Jose, CA

Startrails of Arches by Murali

January 2018



Location/date: Arches National Park, Utah, Jan 05, 2018

OTA info: No telescope, just my camera on a tripod with an intervalometer

Camera/settings: Canon EOS 60d, 17-35mm lens set to 17mm, f2.8, 33.5 minutes total exposure, ISO 800

Lights/Darks: 134 15 second light frames, 1 15 second dark frame

Stacking program: StarStaX on Mac

Processing program: I did not process the images other than stacking them. I had forgotten to take the photos in raw format, so I just kept the photos as is

Anything else: I was with a photographer who helped me set up the shot. He had a light source (LED) that we used to light up the rock formations. A couple of passing cars also helped.



Kid Spot Jokes:

- ♦ **What did the moon say to the therapist?**
I'm just going through a phase!
- ♦ **Why does the moon orbit the earth?**
To get to the other side!
- ♦ **Why did the moon burp?**
Because it was full!

Kid Spot Quiz:



1. **Supermoon occurs during?**
a) low tide b) lunar eclipse c) apogee d) Perigee
2. **How old is the moon?**
a) 4.6 billion years b) 2.6 billion years c) 900 million years
3. **How many astronauts have walked the moon?**
a) 8 b) 22 c) 12

As the Moon Arcs Across the Sky ...

It's not just a time-lapse composite image of our Moon as it arcs across the sky, it's a "super blue blood Moon" eclipse! On Jan. 31, a total lunar eclipse provided a rare opportunity to capture a supermoon, a blood moon and a lunar eclipse all at once. In this composite, the phases of the eclipse form an arc in the sky over NASA Armstrong Flight Research Center's mission support building.

A supermoon occurs when the Moon is closer to Earth in its orbit and appears about 14 percent brighter than usual. As the second full moon of the month, this moon is also commonly known as a blue moon, though it will not be blue in appearance. The super blue moon passed through Earth's shadow and took on a reddish tint, known as a blood moon. This total lunar eclipse occurs when the Sun, Earth and the full Moon form a near-perfect lineup in space and the Moon passes directly behind the Earth into its umbra (shadow).

Image Credit: NASA/Ken Ulbrich



Kid Spot Quiz Answers:

1. Perigee
2. 4.6 billion years
3. 12

From the Board of Directors

Announcements

Spring swap meet will be held on Sunday March 11th from Noon until 3pm. Please visit the website for additional information

Board Meeting Excerpts

December 2, 2017

In attendance

Rob Jaworski, Rob Chapman, Wolf Witt, Bill O'Neil, Vini Carter
Absent Excused: Teruo Utsumi, Swami Nigam, Glenn Newell, Sukhada Palav
Guest: None

In town star party

Proposed policy is that whoever is the host has both the authority to cancel due to the weather and responsibility to email on Wednesday that the event is coming

Looking for an audit firm

SJAA is looking to find a CPA for financial advisor who can do an audit of SJAA's books

Complimentary / Lifetime Memberships

Rob Jaworski proposed that the board consider providing free memberships to active members which would be approved by the board on a case by case basis. No decision was made. A formal set of requirements needs to be developed.

January 6, 2018

In attendance

Bill O'Neil, Sukhada Palav, Glenn Newell, Vini Carter, Wolf Witt, Swami Nigam
Absent Excused: Rob Chapman, Rob Jaworski, Teruo Utsumi
Guest: None

February 3, 2018

In attendance

Teruo Utsumi, Sukhada Palav, Bill O'Neil, Glenn Newell, Vini Carter, Rob Jaworski, Rob Chapman, Swami Nigam
Absent Excused: Wolf Witt

Guest: Ed Wong, Dave Ittner, Emi Nikolov, Nikola Nikolov, Victor Nikolov

Teruo Utsumi resigned from Board and Presidency positions

Teruo Utsumi resigned from Board and Presidency positions. Vice-President Bill O'Neill will temporarily take over the Presidency position until the election next month.

Board members were reelected

Board members who had their term expire were reelected unanimously for another 2 year term. Reelected officers were: Bill O'Neil, Rob Jaworski, Wolf Witt and Glenn Newell

Spring Swap Meet

Dave Ittner will chair the event which will be on March 11th. Bill, Vini and Glenn will be there for full time while Sukhada will be there half time.

Thank you Tom Piller

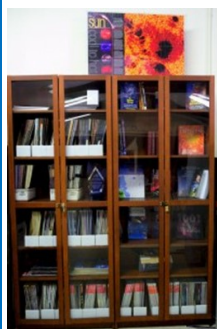
"On behalf of SJAA we would like to thank Tom Piller for his contributions to the Ephemeris as the Production Editor. After four years of producing timely and beautiful editions, Tom has passed the baton to our new Production Editor, Emi Nikolov. Welcome Emi and I look forward to working with you". - Sandy Mohan, Editor

New Production Editor



Emi Nikolov

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick STArT Program

The Quick STArT Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held every other month.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the at-

tendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick STArT program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about. Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Mor-

gan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Vacant
Vice Pres/Dir:	Bill O'Neil
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Rob Chapman
Director:	Vini Carter
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Swami Nigam
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Emi Nikolov
Fix-it Program:	Vini Carter
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Vacant
Loaner Program:	Muditha Kanchana
Memberships:	Dave Ittner
Publicity:	Rob Jaworski
Quick STArT	Dave Ittner
Solar & Starry Nights:	Bill O'Neil
School Events:	Jim Van Nuland
Speakers:	Sukhada Palav
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Place
postage
here

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

- ☐ **New** ☐ **Renewal** (Name only if no corrections)
- ☐ **\$20** Regular Membership with **online** Ephemeris
- ☐ **\$30** Regular Membership with **hardcopy** Ephemeris mailed to below address
- ☐

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____